

Lung ROI Segmentation — SwinUNETR (2D)

1. Task

Voxel-level segmentation of lung foreground (thoracic parenchyma) in axial CT slices. Runs slice-by-slice at native 2D resolution and its outputs are stacked to form the per-series lung bounding box that the downstream nodule model crops to.

- **Input** (1, 256, 256) axial CT slice, intensity-normalised to [0, 1] and resized from the native slice resolution.
- **Output** (1, 256, 256) sigmoid map. Foreground = lung tissue.

2. Model

MONAI SwinUNETR — Shifted-Window (Swin) Transformer encoder + CNN decoder, in 2D. Small-capacity variant chosen to match the parameter budget of the SegResNet baseline (~6.9 M) for an apples-to-apples architecture comparison rather than capacity-vs-capacity.

Property	Value
Architecture family	SwinUNETR (Swin transformer + U-Net)
Spatial dims	2
Input channels	1
Output channels	1 (sigmoid)
feature_size	24 (default is 48; halved for the small variant)
Transformer stage depths	[2, 2, 2, 2]
Attention heads	[3, 6, 12, 24]
Dropout / attn-dropout / path-dropout	0.0
Trainable parameters	6,302,203

3. Data

Unified training corpus assembled from three public sources:

Source	Role in the split
NLST	training + validation
NSCLC-Radiomics	training + validation

Source	Role in the split
LIDC-IDRI	training + validation (lung labels derived from a earlier medium-capacity nodule model)

- **Split:** `unified_v2.json` — patient-grouped, dataset-stratified. 1 683 train / 297 val / 325 test series.
- Slices per split: **365 014 train / 64 097 val / — test held out.**
- **Preprocessing:** each axial slice resized to 256×256 .
- **Class balance:** lung foreground is $\sim 30\text{-}50\%$ of a mid-thorax slice and 0% of pure top-/bottom-of-volume slices — nothing pathological, so plain Dice loss is sufficient.
- **Split parity:** identical split and sampling recipe as the SegResNet baseline; only the model family differs.

4. Training

Setting	Value
Loss	Dice (sigmoid, squared prediction)
Optimizer	Adam
Learning rate	1×10^{-4} (transformer families are more sensitive to LR than CNNs; SegResNet baseline used 1×10^{-3})
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{\text{max}} = 100$, $\eta_{\text{min}} = 1 \times 10^{-6}$
Batch size	16
Samples per epoch	20 000 (random subsample of ~ 365 k train slices)
Epochs (budget)	100
Random seed	42
Mixed precision	bf16 (autocast, no GradScaler)
Hardware	1 $\times$ NVIDIA H100 94 GB
Per-epoch time	≈ 18 min

Best checkpoint (evaluated below): **epoch 59 / 100**, validation Dice 0.9655. Training was still in progress at the time of evaluation; however, val Dice had plateaued in the range $[0.9647, 0.9655]$ for the preceding ~ 15 epochs, so remaining epochs are not expected to move the reported numbers by more than ± 0.001 .

5. Evaluation protocol

Held-out test split (45 751 slices from the `unified_v2` test set). The training loop used the `val` split for early stopping and best-Dice checkpoint selection; the numbers below come from the completely untouched **test** split (never seen during training or model selection). Predictions taken as `sigmoid(logits) > 0.5`. All metrics are computed at the pixel level and micro-averaged over the whole test split ($\sim 3.0 \times 10^9$ pixels).

6. Results

Ticket-required metrics

Metric	Value
mean IoU	0.9829
Accuracy	0.9975
Precision	0.9830
Recall	0.9849

Supplementary

Metric	Value
IoU (foreground / lung class)	0.9684
IoU (background class)	0.9973
Dice / F1 (micro)	0.9839
True positives (px)	227 296 038
False positives (px)	3 937 060
False negatives (px)	3 483 996
True negatives (px)	2 763 620 442

Per-slice Dice distribution (45 751 test slices)

Statistic	Value
Mean	0.9592
Std	0.1240
Min	0.0000
p05	0.8574
Median	0.9888
p95	1.0000
Max	1.0000

Compared to the SegResNet baseline the tails of the per-slice distribution improve noticeably: p05 rises from 0.00 to 0.86 and the per-slice mean jumps from 0.79 to 0.96. In practical terms, the top-/bottom-of-volume slices where the lung is absent or barely present are handled more conservatively (fewer speculative predictions), which is where the SegResNet baseline was losing most of its per-slice score.

7. Comparison with SegResNet baseline

Both models trained on the same split, same augmentation-free pipeline, same batch size, same epoch budget, same eval protocol; both evaluated on the same held-out test set. Only the model family and the LR differ.

Metric	SegResNet (6.9 M)	SwinUNETR (6.3 M)	Δ
mIoU	0.9704	0.9829	+0.0125
Accuracy	0.9957	0.9975	+0.0018
Precision	0.9662	0.9830	+0.0168
Recall	0.9778	0.9849	+0.0071
Dice (micro)	0.9720	0.9839	+0.0119
Dice per slice mean	0.7851	0.9592	+0.1741
Dice per slice p05	0.0000	0.8574	+0.8574

SwinUNETR wins on every reported metric. The largest gain is in the per-slice tail (p05), suggesting the transformer's global receptive field is genuinely useful for the anatomically ambiguous edge slices at the top/bottom of the thorax — which the CNN baseline was mislabelling.

8. Notes on interpretation

Lung ROI segmentation is close to a solved problem on axial CT — the target is a large, contrast-rich region with relatively simple morphology. All four ticket metrics land near their upper bound, which is expected rather than a bug:

- **Accuracy 0.9975** — informative here (unlike the nodule task) because the class ratio is roughly 40 / 60, not 10^{-5} .
- **mIoU 0.9829** — genuinely a mean of two comparable IoUs (0.968 fg, 0.997 bg).
- **Precision 0.9830** — 98 % of predicted-lung pixels are true lung.
- **Recall 0.9849** — 98 % of true-lung pixels are recovered.

Because both models are past the 0.97 mark on the test set, the most operationally meaningful comparison is at the tails of the per-slice distribution (p05) — which is where SwinUNETR shows a real, not just marginal, improvement over the CNN baseline.

9. Reproducibility

- Config `configs/roi_swin.yaml`
- Checkpoint `checkpoints/roi_swin/best_model.pth`
- Metrics JSON `checkpoints/roi_swin/eval_swinunetr_test.json`
- Command `python eval_roi.py --config configs/roi_swin.yaml --ckpt checkpoints/roi_swin/best_model.pth --split test`